

Hybrid Feature-Based Detection of AI-Generated Text

Finn Fonteijn

s2899078

f.f.fonteijn@student.utwente.nl

Linn Gregussen

s3422372

l.y.gregussen@student.utwente.nl

Abstract

1 Introduction

For this project, we wish to explore which linguistic or statistical features are most effective for distinguishing human- vs. AI-generated text. The list of features is not fully determined, but some relevant features might be: sentence structure- and complexity, punctuation and formatting, lexical diversity, perplexity and burstiness. Recent surveys indicate that hybrid approaches, combining statistical and neural methods, often outperform single-modality detectors in robustness and generalization (Wu et al., 2025; Su and Wu, 2024). Thus, we will evaluate individual features as well as combinations and ensembles to identify optimal hybrid configurations.

RQ: Which linguistic or statistical features are most effective in distinguishing human- vs. AI-generated text?

2 Related Work

3 Data

To ensure our model generalizes across diverse English language sources, domains, and LLMs, and to mitigate overfitting to any single dataset's biases, we will utilize multiple datasets for training and evaluation. This approach allows for robust testing of features and hybrids in varied contexts, such as essays, Q&A, and long-form texts from different LLMs.

- AI Text Detection Pile (Hugging Face): ~1.3M samples, diverse texts (Reddit, Wikipedia, books) (Artem9k, 2023).
- AH&AITD (Arslan's Human and AI Text Database): ~12k samples, texts across domains (articles, abstracts, stories, news, reviews) (Akram, 2023).
- LLM - Detect AI Generated Text Dataset (Kaggle sunilthite): >28k essays (Thite, 2023).

- LLM - Detect AI Generated Text (Kaggle Competition): ~10k essays (Kaggle, 2023).
- DAIGT V2 Train Dataset (Kaggle): 48k samples, essays (Deotte, 2023).
- Human ChatGPT Comparison Corpus (HC3): ~24k rows, Q&A from domains (finance, medicine, law, open) (Guo et al., 2023).

4 Method

First the chosen features will be extracted for each entry in the datasets. Examples would be using a language model to calculate perplexity, a tokenizer to calculate lexical diversity or a POS tagger to calculate POS tag ratios. Then, a (lightweight) classifier will be trained on these features; so far logistic regression and XGBoost are candidates. To assess individual and combined efficacy, we will test features separately and explore ensembles (e.g., stacking or voting across feature subsets or classifiers). Lastly, the feature importances will be analyzed.

4.1 Evaluation

The classifier(s) will be evaluated using accuracy, F1-score and AUROC. In addition, precision at different false-positive rates may be reported to mimic realistic screening use.

Depending on the classifier, feature importance will be evaluated using coefficients (for logistic regression), SHAP values (Lundberg, 2017) and/or permutation importance. Hybrid combinations will be benchmarked against baselines to quantify improvements in performance and robustness.

5 Experiment and Results

6 Discussion and Conclusion

References

Arslan Akram. 2023. [AH&AITD: Arslan's Human and AI Text Database](#). Dataset. Associated with the article: An Empirical Study of AI-Generated Text Detection Tools. *Advances in Machine Learning & Artificial Intelligence*, 4(2), 44–55.

- Artem9k. 2023. [Ai text detection pile](#). A dataset for AI text detection tasks, containing human and AI-generated text samples in English. Data fields include TEXT and SOURCE.
- Chris Deotte. 2023. [Daigt v2 train dataset](#). Dataset containing 44k human-written essays and AI-generated ones from different LLMs, including OSS ones.
- Biyang Guo, Xin Zhang, Ziyuan Wang, Minqi Jiang, Jinran Nie, Yuxuan Ding, Jianwei Yue, and Yupeng Wu. 2023. How close is chatgpt to human experts? comparison corpus, evaluation, and detection. *arXiv preprint arxiv:2301.07597*.
- Kaggle. 2023. [Llm - detect ai generated text competition dataset](#). Training dataset contains 1400 samples of human-written and AI-generated text.
- Scott M. Lundberg. 2017. Shap (shapley additive explanations). <https://github.com/shap/shap>. Accessed: 2025-06.
- Yongye Su and Yuqing Wu. 2024. [Robust detection of llm-generated text: A comparative analysis](#). *Preprint*, arXiv:2411.06248.
- Sunil Thite. 2023. [Llm - detect ai generated text dataset](#). Dataset containing more than 28,000 essays written by students and AI-generated.
- Junchao Wu, Shu Yang, Runzhe Zhan, Yulin Yuan, Lidia Sam Chao, and Derek Fai Wong. 2025. [A survey on llm-generated text detection: Necessity, methods, and future directions](#). *Computational Linguistics*, 51(1):275–338.

A Example appendix